Contractibility of the space of connected-truncated factorizations

The Univalent Foundations Program

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Editorial note. The task keeps the complete author proofs of the image and fibre-comparison lemmas and of the final theorem. No exercise is itself counted as a proved problem.

Editorial note. In the explanatory sentence1502,n-truncated for g_1 refers to the already statedn-connected hypothesis1460; the surrounding contraction of the truncated fibre fixes that immediate adjective slip.

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Allowed standard prerequisites (editorial).

- Explicit univalent foundations boundary. Univalence, dependent sums and path transport, n-truncation, n-connected maps and the standard equivalence between iterated homotopy fibres and the fibre of a composite. The displayed exercise pointer for that elementary fibre identity is an explicitly allowed prerequisite, not an extracted exercise solution.

Source reference. Original corollary 7.5.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 187–187. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: cor.

Source reference. Original lemma 7.5.13: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1571–1571. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

Source reference. Original corollary 7.3.10: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1873–1873. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: cor.

Source reference. Original exercise 4.4: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1414–1414. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: ex.

Source reference. Original lemma 4.8.2: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1892–1892. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

Source reference. Original lemma 4.2.5: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1533–1533. Original book reference to explicitly allowed type-theoretic foundations or unselected surrounding discussion. The counted local construction is retained; no external pointer supplies the target theorem. Original reference category: lem.

We can now construct the factorization, in a fairly obvious way.

Definition 7.6.1. Let $f : A \to B$ be a function. The **n-image** of f is defined as

$$\mathsf{im}_n(f) :\equiv \sum_{b:B} \|\mathsf{fib}_f(b)\|_n.$$

When $n = -1$, we write simply $\mathsf{im}(f)$ and call it the **image** of f.

Lemma 7.6.2. *For any function $f : A \to B$, the canonical function $\tilde{f} : A \to \mathsf{im}_n(f)$ is n-connected. Consequently, any function factors as an n-connected function followed by an n-truncated function.*

Proof. Note that $A \simeq \sum_{(b:B)} \mathsf{fib}_f(b)$. The function $\tilde{f}$ is the function on total spaces induced by the canonical fiberwise transformation

$$\prod_{b:B} \Big(\mathsf{fib}_f(b) \to \|\mathsf{fib}_f(b)\|_n\Big).$$

Since each map $\mathsf{fib}_f(b) \to \|\mathsf{fib}_f(b)\|_n$ is n-connected by original corollary 7.5.8, $\tilde{f}$ is n-connected by original lemma 7.5.13. Finally, the projection $\mathsf{pr}_1 : \mathsf{im}_n(f) \to B$ is n-truncated, since its fibers are equivalent to the n-truncations of the fibers of f. □

In the following lemma we set up some machinery to prove the unique factorization theorem.

Lemma 7.6.3. *Suppose we have a commutative diagram of functions*

$$\begin{array}{ccc} A & \xrightarrow{g_1} & X_1 \\ {\scriptstyle g_2}\downarrow & & \downarrow{\scriptstyle h_1} \\ X_2 & \xrightarrow[h_2]{} & B \end{array}$$

with $H : h_1 \circ g_1 \sim h_2 \circ g_2$, where g_1 and g_2 are n-connected and where h_1 and h_2 are n-truncated. Then there is an equivalence

$$E(H, b) : \mathsf{fib}_{h_1}(b) \simeq \mathsf{fib}_{h_2}(b)$$

for any $b : B$, such that for any $a : A$ we have an identification

$$\overline{E}(H, a) : E(H, h_1(g_1(a)))(g_1(a), \mathsf{refl}_{h_1(g_1(a))}) = (g_2(a), H(a)^{-1}).$$

Proof. Let $b : B$. Then we have the following equivalences:

$$\begin{aligned}
\mathsf{fib}_{h_1}(b) &\simeq \sum_{w:\mathsf{fib}_{h_1}(b)} \|\mathsf{fib}_{g_1}(\mathsf{pr}_1 w)\|_n && \text{(since } g_1 \text{ is } n\text{-connected)} \\
&\simeq \Big\| \sum_{w:\mathsf{fib}_{h_1}(b)} \mathsf{fib}_{g_1}(\mathsf{pr}_1 w) \Big\|_n \\
&&& \text{(by original corollary 7.3.10, since } h_1 \text{ is } n\text{-truncated)} \\
&\simeq \|\mathsf{fib}_{h_1 \circ g_1}(b)\|_n && \text{(by original exercise 4.4)}
\end{aligned}$$

and likewise for h_2 and g_2. Also, since we have a homotopy $H : h_1 \circ g_1 \sim h_2 \circ g_2$, there is an obvious equivalence $\mathsf{fib}_{h_1 \circ g_1}(b) \simeq \mathsf{fib}_{h_2 \circ g_2}(b)$. Hence we obtain

$$\mathsf{fib}_{h_1}(b) \simeq \mathsf{fib}_{h_2}(b)$$

for any $b : B$. By analyzing the underlying functions, we get the following representation of what happens to the element $(g_1(a), \mathsf{refl}_{h_1(g_1(a))})$ after applying each of the equivalences of which E is composed. Some of the identifications are definitional, but others (marked with a $=$ below) are only propositional; putting them together we obtain $\overline{E}(H, a)$.

$$\begin{aligned}
(g_1(a), \mathsf{refl}_{h_1(g_1(a))}) &\overset{=}{\mapsto} \big((g_1(a), \mathsf{refl}_{h_1(g_1(a))}), \big|(a, \mathsf{refl}_{g_1(a)})\big|_n\big) \\
&\mapsto \big|((g_1(a), \mathsf{refl}_{h_1(g_1(a))}), (a, \mathsf{refl}_{g_1(a)}))\big|_n \\
&\mapsto \big|(a, \mathsf{refl}_{h_1(g_1(a))})\big|_n \\
&\overset{=}{\mapsto} \big|(a, H(a)^{-1})\big|_n \\
&\mapsto \big|((g_2(a), H(a)^{-1}), (a, \mathsf{refl}_{g_2(a)}))\big|_n \\
&\mapsto \Big((g_2(a), H(a)^{-1}), \big|(a, \mathsf{refl}_{g_2(a)})\big|_n\Big) \\
&\mapsto (g_2(a), H(a)^{-1})
\end{aligned}$$

The first equality is because for general b, the map $\mathsf{fib}_{h_1}(b) \to \sum_{(w:\mathsf{fib}_{h_1}(b))} \|\mathsf{fib}_{g_1}(\mathsf{pr}_1 w)\|_n$ inserts the center of contraction for $\|\mathsf{fib}_{g_1}(\mathsf{pr}_1 w)\|_n$ supplied by the assumption that g_1 is n-truncated; whereas in the case in question this type has the obvious inhabitant $\big|(a, \mathsf{refl}_{g_1(a)})\big|_n$, which by contractibility must be equal to the center. The second propositional equality is because the equivalence $\mathsf{fib}_{h_1 \circ g_1}(b) \simeq \mathsf{fib}_{h_2 \circ g_2}(b)$ concatenates the second components with $H(a)^{-1}$, and we have $H(a)^{-1} \cdot \mathsf{refl} = H(a)^{-1}$. The reader may check that the other equalities are definitional (assuming a reasonable solution to original exercise 4.4). □

Theorem 7.6.4. *For each $f : A \to B$, the space $\mathsf{fact}_n(f)$ defined by*

$$\sum_{(X:\mathcal{U})} \sum_{(g:A\to X)} \sum_{(h:X\to B)} (h \circ g \sim f) \times \mathsf{conn}_n(g) \times \mathsf{trunc}_n(h)$$

is contractible. Its center of contraction is the element

$$(\mathsf{im}_n(f), \tilde{f}, \mathsf{pr}_1, \theta, \varphi, \psi) : \mathsf{fact}_n(f)$$

arising from Lemma 7.6.2, where $\theta : \mathsf{pr}_1 \circ \tilde{f} \sim f$ is the canonical homotopy, where φ is the proof of Lemma 7.6.2, and where ψ is the obvious proof that $\mathsf{pr}_1 : \mathsf{im}_n(f) \to B$ has n-truncated fibers.

Proof. By Lemma 7.6.2 we know that there is an element of $\mathsf{fact}_n(f)$, hence it is enough to show that $\mathsf{fact}_n(f)$ is a mere proposition. Suppose we have two n-factorizations

$$(X_1, g_1, h_1, H_1, \varphi_1, \psi_1) \qquad \text{and} \qquad (X_2, g_2, h_2, H_2, \varphi_2, \psi_2)$$

of f. Then we have the pointwise-concatenated homotopy

$$H :\equiv (\lambda a.\, H_1(a) \cdot H_2^{-1}(a)) \;:\; (h_1 \circ g_1 \sim h_2 \circ g_2).$$

By univalence and the characterization of paths and transport in Σ-types, function types, and path types, it suffices to show that

(i) there is an equivalence $e : X_1 \simeq X_2$,
(ii) there is a homotopy $\zeta : e \circ g_1 \sim g_2$,
(iii) there is a homotopy $\eta : h_2 \circ e \sim h_1$,
(iv) for any $a : A$ we have $\mathsf{ap}_{h_2}(\zeta(a))^{-1} \cdot \eta(g_1(a)) \cdot H_1(a) = H_2(a)$.

We prove these four assertions in that order.

(i) By Lemma 7.6.3, we have a fiberwise equivalence

$$E(H) : \prod_{b:B} \mathsf{fib}_{h_1}(b) \simeq \mathsf{fib}_{h_2}(b).$$

This induces an equivalence of total spaces, i.e. we have

$$\Big(\sum_{b:B} \mathsf{fib}_{h_1}(b)\Big) \;\simeq\; \Big(\sum_{b:B} \mathsf{fib}_{h_2}(b)\Big).$$

Of course, we also have the equivalences $X_1 \simeq \sum_{(b:B)} \mathsf{fib}_{h_1}(b)$ and $X_2 \simeq \sum_{(b:B)} \mathsf{fib}_{h_2}(b)$ from original lemma 4.8.2. This gives us our equivalence $e : X_1 \simeq X_2$; the reader may verify that the underlying function of e is given by

$$e(x) \equiv \mathsf{pr}_1(E(H, h_1(x))(x, \mathsf{refl}_{h_1(x)})).$$

(ii) By Lemma 7.6.3, we may choose $\zeta(a) :\equiv \mathsf{ap}_{\mathsf{pr}_1}(\overline{E}(H, a)) : e(g_1(a)) = g_2(a)$.

(iii) For every $x : X_1$, we have

$$\mathsf{pr}_2(E(H, h_1(x))(x, \mathsf{refl}_{h_1(x)})) : h_2(e(x)) = h_1(x),$$

giving us a homotopy $\eta : h_2 \circ e \sim h_1$.

(iv) By the characterization of paths in fibers (original lemma 4.2.5), the path $\overline{E}(H, a)$ from Lemma 7.6.3 gives us $\eta(g_1(a)) = \mathsf{ap}_{h_2}(\zeta(a)) \cdot H(a)^{-1}$. The desired equality follows by substituting the definition of H and rearranging paths. □